Supplementary materials for: Palaeoclimatic constraints behind the Great American Biotic Interchange: insights from eco-evolutionary simulations

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Method details

Beforehand, it has to be noticed that, across the entire study, "North America" actually refers to the current North and Central American continents. The delimitation between North and South America was arbitrarily set as a straight line connecting two points located along the eastern and western coasts of present Panama (−82.882°lon, 7.138°lat and −81.369°lon and 9.008°lat) (see main text, Figure 1).

Palaeoclimate data

We used the spatio-temporal terrestrial palaeoclimate series from the PALEO-PGEM atmosphere-ocean general circulation model [1], covering the last 5 Myrs with a 1 × 1 ° spatial resolution [2]. For the scope of this work, we used two bioclimatic variables: mean annual temperature (MAT, BIO1) and mean annual precipitation (MAP, BIO12). To avoid reaching an overkillingly-high number of simulation steps, we reduced the temporal resolution of the climate simulations from a 1000-years to a 10,000-years time step.

Additionally, in order to discriminate simulations based on the biome where the initial species originates, we reclassified monthly palaeoclimate layers for our starting time (t = 5 Myrs) according to Köppen-Geiger broad climate types [3,4]. This classification divides the world into five broad climatic regions: Polar, Cold, Temperate, Arid and Tropical, following the criteria proposed by Köppen [5].

Simulation model and modelling schemes

We performed spatially-explicit simulations of biodiversity using the General Engine for Eco-Evolutionary SImulationS, gen3sis, v1.5.11 [6]. Simulations, moving forward-in-time with a 10ky time step, tracked the radiation of a clade arising from a single ancestral population, either originating in North or in South America, throughout the last 5 Myr. In gen3sis, virtual species are modelled with individual populations evolving independently, following a set of four rules: dispersal, speciation, environmental filtering and trait evolution. All simulation scenarios that we implemented only differed on the last rule (trait evolution), the first three being common to all models (or at least partly).

We elaborated a gridded landscape relying on the two time-series of the aforementioned palaeoclimatic layers. Within this landscape, populations making up virtual species inhabited a set of connected cells. At each time step, inter-cell dispersal could occur with an intensity sampled in a Weibull distribution (of shape and scale respectively termed ϕ and ψ). If, by chance, two populations happened to be disconnected, they would accumulate divergence as long as they get disconnected, at a rate ruled by a divergence factor, here set to 0.05 (Table 1). Once passed a divergence threshold (S), speciation would occur – note that, consequently, the only mode of speciation allowed by gen3sis is allopatry.

As our goal was to evaluate the extent to which palaeoclimate change may have shaped the unequal success of North American taxa establishing southwards compared to South American taxa establishing northwards, local climate was set as the only constraint for a population to maintain in a given cell. Each species was therefore assigned a two-dimensional climatic niche, one axis corresponding to temperature suitability and the other to precipitation suitability. The abundance of a species along an environmental gradient was ruled by two parameters – that we also call traits – : a population- and species-specific optimum (resp. $P_{k,l}$ and $T_{k,l}$ for the precipitation and temperature optima of the population k of the species l) and a breadth (or tolerance) (resp. ω_P and ω_T; within a simulation, breadths were shared across populations and species). A population could maintain in a given cell only if the difference between its optima and local climate (both temperature and precipitation) was below its breadths. Therefore, given a site i with local precipitation and temperature at time t being $P_i(t)$ and $T_i(t)$, this translates in

$$\begin{cases} |P_i(t) - P_{k,l}| < \omega_P \\ |T_i(t) - T_{k,l}| < \omega_T \end{cases}$$

Each simulation started with a single ancestral species. The range of this ancestral species was randomly selected within the desired ancestral area (*i.e.*, North or South America). To avoid oversampling high latitudes, we merged the ancestral landscape with the centroids of an equal-area grid (~100km spacing) generated using the R package *h3jsr* [7]. We then sampled a centroid at random and retrieved the corresponding cell in our landscape. To reduce the probability of initial crash, the climate optima parameters of the ancestor were set to those of the local environment. However, due to the geometry of the two continents, this sampling scheme resulted in ancestral species starting from South America originating on average closer to the isthmus of Panama than those from North America (Figure 1 A). To control for this spatial bias, we assessed the distance to the isthmus of each cell of the

North American portion of our $1 \times 1°$ landscape. We then carried out other batches of simulations starting from North America, constraining the initial distance to the isthmus of the ancestral species of the simulation i to be as close as possible to same distance involving the ancestral species of the simulation i starting from South America (Figure 1 B).

500 replicates were made under each modelling scheme (detailed in the next paragraph), both with North and South America as ancestral area. Across replicates, we varied 6 key parameters: (1) the divergence threshold above which speciation occurred (S, range = $[1, 3]$), (2) the rate of environmental niche evolution – only in models where we allowed climatic niche to evolve (σ_e, $[0.005, 0.15]$), (3) the shape (ϕ, $[0.5, 3]$) and (4) scale (ψ, $[0.1, 1]$) of the Weibull distribution acting as a dispersal kernel, (5) the strength of environmental filtering (*i.e.*, niche breadth) for temperature (ω_T, $[0.01, 0.1]$) and (6) precipitation (ω_P, $[0.05, 0.3]$). As done by many authors, we ensured an even sampling of the resulting multidimensional parameter space via a quasirandom sampling technique relying on Sobol' sequences [8,9]. Parameter ranges were determined on the basis of a literature review, allowing to assess their biological relevance, and were further adjusted in order to limit the prevalence of crashing simulations. In addition to these varying parameters, we specified the value of several fixed parameters (Table 1)

We implemented various simulation scenarios where we modulated the ability of species to cope with climate change, and thus, the generalist vs. specialist nature of the simulated species. In the full-adaptationist model (*M1*, generalist), climatic optima ($P_{k,l}$ and $T_{k,l}$) were able to evolve in the direction of local climate change. To isolate the effect of a single environmental covariate on the resulting simulated biodiversity pattern, alternative versions of this model consisted of either removing the temperature (*M2*) or the precipitation (*M3*) constraint on species ecology. Besides, we also implemented a zero-adaptationist model (*M0*, specialist), where niche optima could not evolve and were therefore held constant (to their initial value) across all simulations.

Last, it is important to notice that, since the palaeoclimate reconstructions that we used ignored tectonic plate motion – a reasonable assumption for a 5-Myr time frame – continental position in our landscapes was held constant across simulations. Therefore, in that framework, changes in connectivity between North and South America could only arise from the environment.

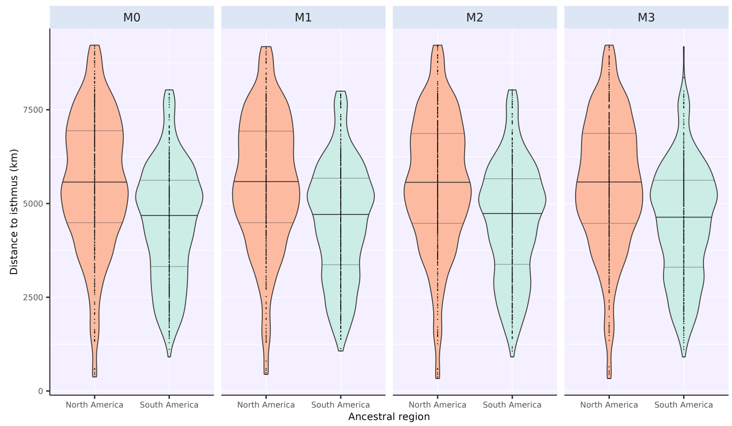
Postprocessing and quantifications

At every time step, for each run of each simulation, we used a function released by Skeels et al. [9] to save presence-absence matrices. Given a time t, those matrices stored on their columns the presence of each simulated species that have existed from the onset of the simulation to t within each cell of the landscape (rows of the matrix) as a binary variable – 1: the species inhabits that cell, 0: it does not.

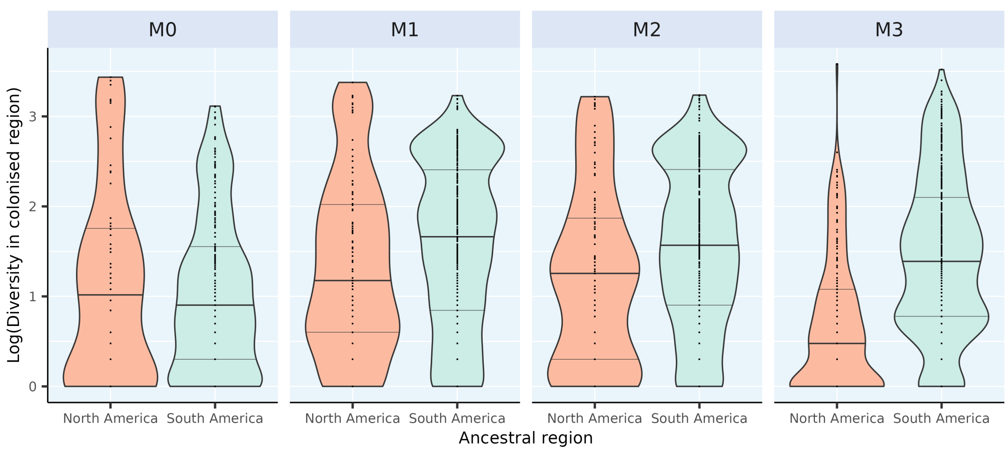
Given a batch of simulations, after discarding those that crashed – *i.e.*, that lead to no living representative – we quantified four metrics reflecting the occurrence, magnitude and efficiency of an eventual transcontinental exchange based on the presence-absence matrix of the last time step of each successful run (hereafter called t_{end}). First, we assessed the exchange success, a binary variable set to one if at least one pixel of the area to colonise (hereafter called $\mathcal{A}_{col}$, *e.g.*, South America if the simulation started in North America) recorded a presence. Then, we recorded the biome in which the initial ancestor occurs (*i.e.*, Tropical, Arid, Temperate, Cold, Polar) as well as the distance of this initial ancestor to a point arbitrarily located on the zone of the isthmus of Panama, ($-84°$lon, $10°$lat). Last, in the case of a successful exchange, we quantified the number of virtual taxa recorded in $\mathcal{A}_{col}$ at t_{end} and the proportion of $\mathcal{A}_{col}$ actually colonised at t_{end} as the ratio between the number of cells occupied by the colonisers and the total number of available cells (*i.e.*, surface of $\mathcal{A}_{col}$).

Supplementary figures

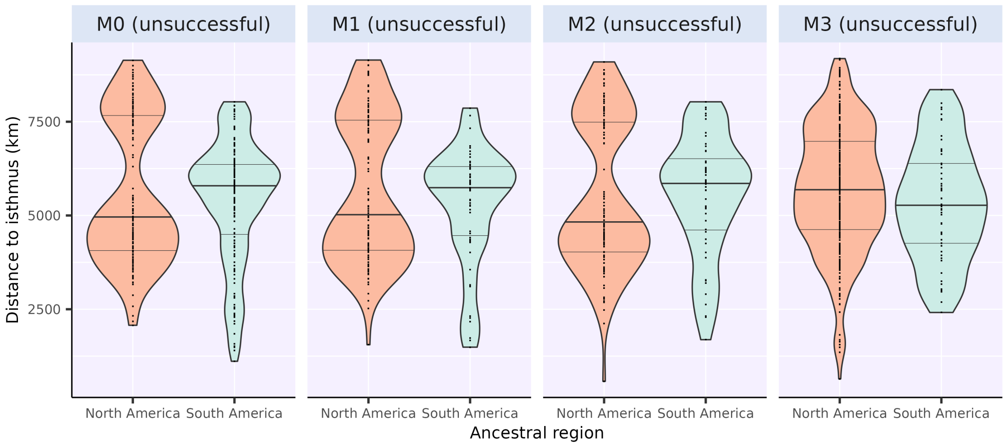
All across the study, metrics computed out of simulations starting in North America are shown in coral red, and those starting in South America in cyan.



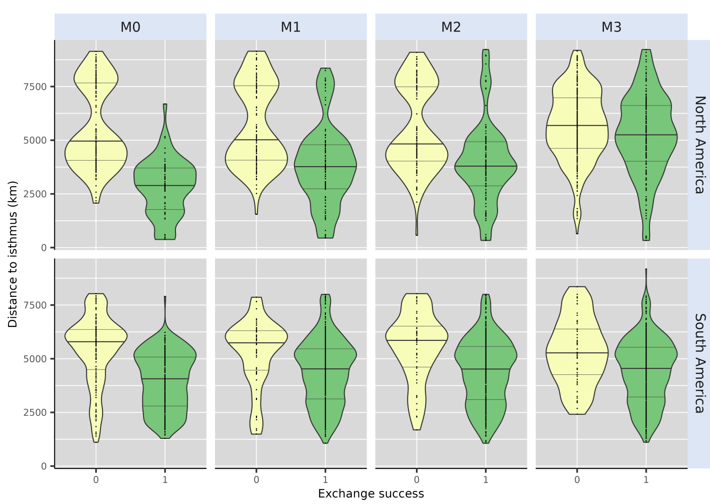
Supp.Figure 1: Distribution of the distance to isthmus of the ancestral species of simulations either starting in North or South America, resulting from our random equal-area sampling scheme (see Method details). Dashed lines on top and bottom of the distribution represent the 25% and 75% quantiles whereas the thicker full line in the middle shows the median (50% quantile).



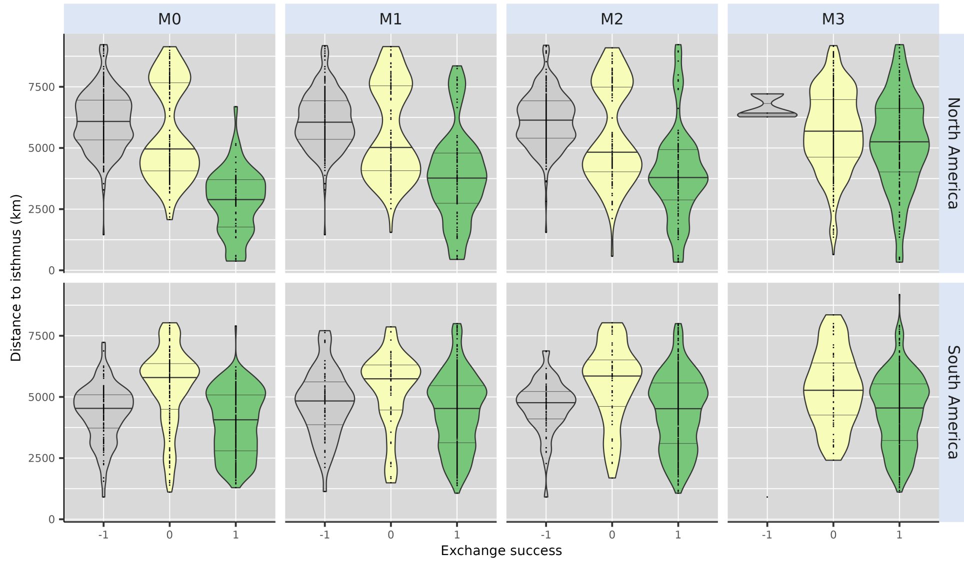
Supp.Figure 2: Diversity in the colonised area following a successful interchange (decimal log scale). Dashed lines on top and bottom of the distribution represent the 25% and 75% quantiles whereas the thicker full line in the middle shows the median (50% quantile).



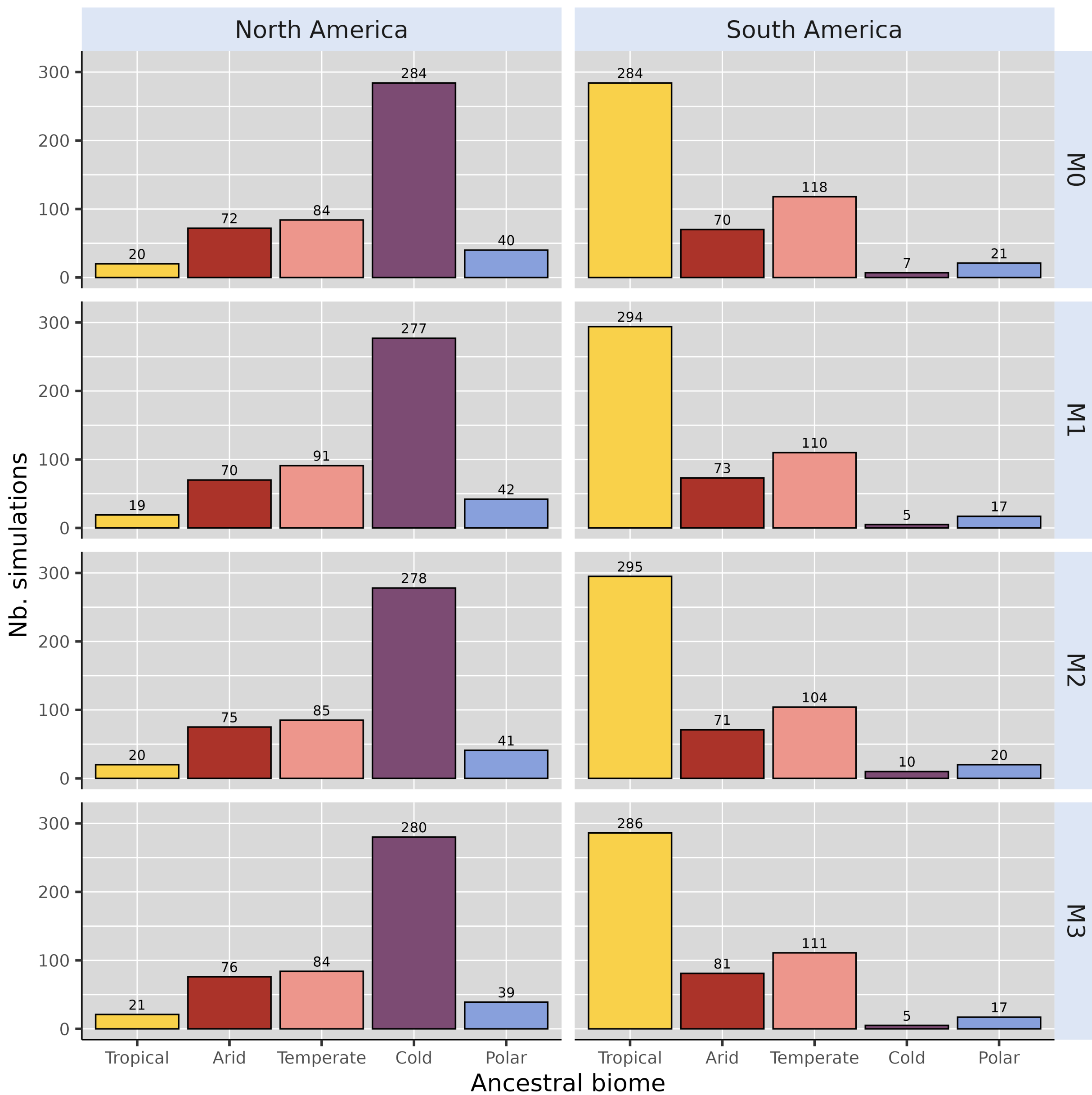
Supp.Figure 3: Distance to the isthmus for non-successful simulations – *i.e.*, that did not result in a significant interchange. Dashed lines on top and bottom of the distribution represent the 25% and 75% quantiles whereas the thicker full line in the middle shows the median (50% quantile).



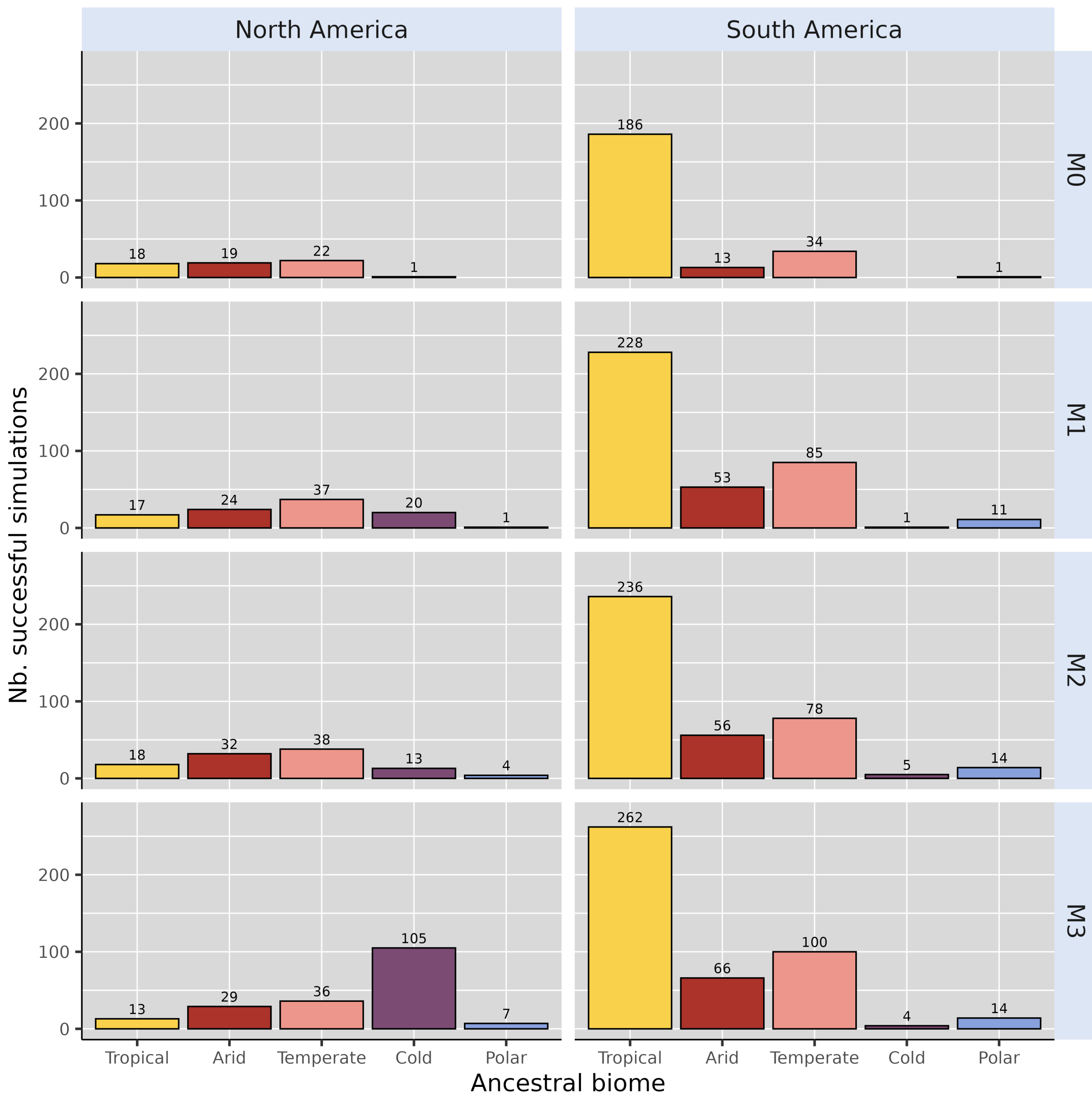
Supp.Figure 4: Comparison of the initial distance to the isthmus between successful and unsuccessful simulations of each of our four models, either starting from North (top) or South (bottom) America. Successful simulations ($Exchange\ Sucess = 1$) are displayed in green, whereas unsuccessful simulations ($Exchange\ Sucess = 0$) are displayed in yellow. Dashed lines on top and bottom of the distribution represent the 25% and 75% quantiles whereas the thicker full line in the middle shows the median (50% quantile).



Supp.Figure 5: Comparison of the initial distance to the isthmus between successful and unsuccessful simulations of each of our four models including simulations that crashed (*i.e.*, resulted in no living representative at the final time step), either starting from North (top) or South (bottom) America. Successful simulations ($Exchange\ Sucess = 1$) are displayed in green, unsuccessful simulations ($Exchange\ Sucess = 0$) in yellow and simulation that crashed ($Exchange\ Sucess = -1$) in grey. Dashed lines on top and bottom of the distribution represent the 25% and 75% quantiles whereas the thicker full line in the middle shows the median (50% quantile).



Supp.Figure 6: Biome where the ancestral species of each of the 500 simulation originated, for each continent and each scenario.



Supp.Figure 7: Biome where the ancestral species of each simulation leading to a successful exchange originated, for each continent and each scenario.

Supplementary tables

Supp.Table 1: Fixed parameter values for gen3sis simulations.

Name	Value	Description
`divergence_factor`	0.05	How much divergence disconnected populations accumulate across iterations
`max_species`	10000.00	Maximum number of simulated species a simulation can handle (if falls above, the simulation crashes)
`max_coex_species`	1000.00	Cell carrying capacity
`start_species`	1.00	Number of species to start with
`init_ab`	500.00	Initial abundance of the start species
`grid_cell_distance`	111.00	Distance (in km) between two grid cells of the landscape at the equator

Supplementary references

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